Flutter Architecture v2.1

Compatibility addendum to the existing 13-page Flutter Architecture v2.0 | 10 Nov 2026 | Proposed, not signed-off

Source precedence

Use Master BRD v2.1 for updated voucher intake and unresolved debit milestone; Admin Working Process v1.1 for Admin voucher upload and fulfillment; Flutter Architecture v2.0 for existing UI, routes, maps, themes, feature boundaries and navigation. This page updates only the affected Flutter responsibilities; the original 13 pages follow unchanged.

Mobile Wallet and voucher UI - what changes

Scope	Frontend responsibility
Wallet / Redemption	Display posted/total, reserved and spendable Coins separately. Submitting a redemption creates a pending request with a hold; do not show a permanent debit before the backend confirms the configured settlement milestone.
Request status	Show Pending / Under Review / Approved / Rejected / Fulfilled as returned by the backend. Approval is not the same as an issued, spendable voucher.
My Vouchers	Display real issued/assigned voucher details only after the server authorizes assignment. Show expiry, merchant and amount; never fabricate QR/code.
Admin voucher intake	Image/PDF upload, OCR and barcode detection belong to Angular Admin and backend processing, not to a new Flutter consumer capture flow.
Optional OCR / QR	OCR and barcode scanning remain PROPOSED, not approved for the initial release. Flutter can display issued voucher QR only if provided securely.

Open decision VCH-DEC-01

Do not assume Coins are finally debited on Admin Approve OR on Fulfill. Master BRD and Admin working process differ. Backend will own the authoritative state; mobile reacts to returned posted/reserved/spendable balances and voucher state. Confirm the debit milestone before implementing the final wallet state machine.

Not changed

No changes to Traveler, Hunter, Game, Friends, profile navigation, Maps redesign, styling or main six UI flows from the v2.0 architecture. HP/RC/XP remain retired; Coins-only is maintained.

Document control: this is an additive compatibility note, not confirmation that Flutter API integration is already implemented.

WAYHUNTER / ENGINEERING BLUEPRINT

Flutter Frontend Architecture v2.0

புதிய UI Screens, முழு BRD விதிகள் மற்றும் தற்போதைய GitHub Code அடிப்படையில் — புதிய Map View மற்றும் Theme Redesign உடன்.

Feature-First | Clean Architecture | MVVM + Riverpod | Real Maps | Coins Only

42 Screens Reviewed · 7 Screenshot Groups
24-page Master BRD v2.0 · Business Logic
GitHub Existing Flutter · Non-destructive migration
Map-first Redesign · Traveler / Hunter / Game

10 NOVEMBER 2026

PROPOSED ARCHITECTURE • REPOSITORY UNCHANGED

1. தேர்ந்தெடுக்கப்பட்ட Architecture

Feature-First + Clean Architecture + MVVM + Riverpod 3 + GoRouter + Repository Pattern (existing approach retained). Build Android-first Flutter mobile UI and call **one ASP.NET Core backend** also used by Angular Admin. `data → domain → presentation` boundaries should exist per feature, but existing Flutter screens and route names must remain stable during refactoring.

```
UI Screen (Flutter widgets)
    ↓ actions / observes immutable state
Riverpod Notifier / AsyncNotifier (ViewModel)
    ↓ feature use-case when needed
Domain Entity + Repository Contract
    ↓
Data Repository + API DataSource / local draft cache
    ↓ HTTPS / SignalR
ASP.NET Core Wayhunter API → EF Core / SQL Server / Cloudinary / integrations
```

Backend is authoritative: registration/UserId, OTP, Hunter/Game approvals, actual Start/Finish Trip, Coins, voucher issuance, leaderboards, hazard validity. Flutter may show optimistic *pending* status when an API request is accepted, but it cannot self-approve/self-credit. Real-time status is a convenience, not a replacement for API refetch.

2. Audit: ZIP-இல் உண்மையாக உள்ள Screen References

The uploaded ZIP contains **42 image files**, not the complete final BRD screen inventory. These are screenshots of an existing prototype, including DEMO labels. They are **visual references**; they do not prove backend connectivity.

Group	Images checked	Example screens	Main difference from final BRD
Login & Home	4	login, create, forgot, home	DEMO login; OTP, reset and permissions not shown in ZIP
Traveler	10	traveler map, create trip, active trip, find, hazard, memory	Map unavailable, demo place results, straight-line routes; global leaderboard visible
Hunter	5	home, add place, 360 camera, 360 done, review	Demo GPS/maps/evidence; pending/admin flow incomplete in pictures
Game	14	game zone, create, hint, play, proof, My Games	HP/RC text remains; district map and old leaderboard; manual approvals and expiry must be reflected
Friends	4	friends hub, discover, invite, contacts	Invitation is demo link; contacts are demo matches
Profile	1	profile home	Other 17 profile/settings views absent from ZIP
Coins Wallet	4	wallet home, redeem, vouchers, rules	Visual Coins flow exists; real redemption/QR and backend approval not connected

What to preserve visually: hierarchy, 390×844-oriented layouts, key input placements, screen meaning, brand identity, legible glass cards and action order. **What may change:** backgrounds, overlay contrast, accent

colors, map container/layout, map controls, demo notices, obsolete rewards, outdated bottom-nav buttons. Do **not** paste screenshot PNGs as full-screen UI; build native responsive Flutter widgets.

Direct visual findings

- `traveller/travellerhome.png`: huge map preview says **Map unavailable** behind two large actions. Replace with a live Sri Lanka map; preserve **Create New Trip** and **My Plans** access.
- `traveller/start trip.png`: map area says **Google Maps needs Android configuration**. Replace with full-width live trip map, trip progress, route line, Find/Hazard/Memory/Friends tools, verified hazard badges.
- `traveller/find.png`: results list is explicitly **local DEMO**. Replace with on-demand 1-km provider search and map/list toggle.
- `traveller/createnewtrip.png`: long form and route builder are valuable, but place selection must use searchable real map and ordered stops with road route preview.
- `hunter/hunderhome.png` and `hunter/addlocation.png`: demo district and map. Map must display permitted district and **approved published places** only; GPS accuracy/restriction checks still enforced by backend.
- `game/playgame2.png`: Sri Lanka district overview can remain **optional** as a directory selector, but must not disclose hidden answer locations or become a compulsory GPS/km clue.
- `game/create game.png`, `game/game from.png` and `game/hint&rule.png` show obsolete **HP/RC**; new screens display **Coins-only** and correct approval status.
- `wallet/home.png` already has 2,000-Coin star/bar, and `wallet/rewards.png` shows 10,000 Coins = Rs.100; retain layout, replace local demo authority with secure API source.

3. Navigation: six requested flows + cross-feature links

Flow	Root	Screens / important transitions	Architecture owner
1 — Authentication → Home	`/sign-in`	Register → OTP → verified session → permissions as needed → Home; forgot/reset, logout	`features/auth`, `features/home`
2 — Home → Traveler	`/traveller`	Landing map → Create Trip/ Place Picker → Suggested/AI/ Reuse → My Plans → Start → Live Map → Nearby/Hazard/ Memories/Friends → Finish → History/Leaderboard	`features/traveller` + `features/friends`, map/ safety services
3 — Home → Hunter	`/hunter`	GPS district → New Place → Live guided 360 → Preview → Live photo → Duplicate check → Submit → My Hunts Pending/Approved/Rejected → module leaderboard	`features/hunter`
4 — Home → Game	`/game`	Create Draft → Photo/Text Hints → Admin Pending → Active/Expired; Play → Hints → Live Proof → Admin result → History/Leaderboard	`features/games`
5 — Profile → Settings	`/profile`	Profile five tabs → unified Friends → Edit/Language/ Privacy/Notifications/Security/ Delete/Logout	`features/profile`, `features/friends`

Flow	Root	Screens / important transitions	Architecture owner
6 — Profile → Wallet	`/wallet`	Coins dashboard → Redeem → Confirm → Pending → Approved/Rejected → My Vouchers/History/Rules	`features/wallet`

Routes: Keep existing `lib/app/app_router.dart`, `features/*/presentation/providers/*_routes.dart` and stable URLs. Add `leaderboards/` module destinations, `notifications/` routes and typed route parameters in additive steps. `Profile → Followers/Following` and `Create Trip → Add Friends` open **the same Friends hub** with different `openedFrom` context; when returning, preserve the unsent Trip Draft. Deep links must be sanitized, and Auth Guards should protect private routes.

BRD navigation conflict: global Leaderboard must be hidden/disabled on Home, Profile, Friends and Wallet. The replacement of the old fourth global bottom-nav slot remains an **Owner decision**. Until approved, a clearly usable **Home / Traveler / Game / Profile** four-tab pattern (as pictured on wallet) is a **suggestion**, not a settled BRD fact. Hunter and three leaderboards must still be directly accessible from their respective module pages.

4. Map Redesign: all major screens

4.1 Map System Architecture (shared)

```
features/traveller / hunter / places / weather / notifications
           ↓ view-specific Map Presentation
core/maps/MapFacade + MapControllerAdapter
           ├── Google Maps SDK (live Android tiles, markers, camera)
           ├── PlacesSearchRepository (Wayhunter backend / approved Places provider)
           ├── RoutesRepository (backend proxied road Directions/Routes)
           ├── LiveLocationRepository (Geolocator + permission / accuracy)
           ├── DistrictBoundaryRepository (approved polygon data)
           ├── ApprovedPlacesRepository (approved public places)
           └── HazardOverlayRepository (active nonexpired reports)
```

The map component is not the location source of truth. Use `LatLng/GeoPoint`, selected Place ID, source, accuracy, timestamps, typed markers. Only a *route service response* may be labeled driving/walking road route and distance/ETA. Straight-line polyline must be labelled preview only; never pass it off as actual roads.

4.2 Required screen-specific map behaviors

Screen	New map UI	Primary actions and rules
T-01 Traveler Home	**Large actual Sri Lanka map** filling available content, rounded/clipped responsive edges, custom trusted-place markers, accessible recenter/search	Create Trip, My Plans fixed and visible; map failure shows retry, not a fake background map
T-03 Trip Place Picker	Full map with search field, autocomplete list, selected pin, compact place preview bottom sheet	Set Start / Add Stop / Set End; manually search when GPS denied; return PlaceId + name + coordinates
T-02 Create Trip / Route Builder	Ordered stop chips + expandable real route preview, travel mode, reposition/ reorder controls	Show road routes only after Directions API result; no false ETA; protect draft during picker/friends flow
T-06 Active Trip Live Map	**Map-first full screen** with route, current user location, consenting friends, next stop and hazard icons; floating recenter/ navigation	Explicit Start/Pause/Finish; Find, Report, Memories, Friends; end sharing on finish

Screen	New map UI	Primary actions and rules
T-07 Find Nearby	Map/list toggle with categories and result bottom sheet; 1-km radius when user taps Find	Food/hospital/fuel/police/ATM/ attractions; no continuous background search
T-08 Hazard + Route Alert	Nearby valid hazards marked by severity, freshness and verification state, dismissible banner/pop-up	Pop-up only while actually on trip, on matching route/time/direction; do not imply unverified is official
H-01 Hunter Home	Current district region and approved public discovery pins; search/list toggle	User can submit in **detected district** only; no pending/rejected places on public map
H-02 GPS Evidence	Map mini-preview + precise status/ accuracy and restricted-area badge	Camera-only evidence and 200-m duplicate *candidate* check; no user coordinate override
G-Play District Selection	Optional Sri Lanka district **selector**, no secret stage pin or km meter	Game clues remain photo + text + timer; removing district map must not break game play
Weather Overlay	Optional on Traveler map when authorized/provider supports it	Show provider, timestamp, legend, unavailable state; do not conflate weather with community hazards

4.3 Map UI layers & visual specification (recommended)

- **Base layer:** genuine Google Map on Android, `GoogleMap` widget properly bounded and lifecycle-aware. Apply suitable map styling; do not cover map tiles with a scenic photo.
- **Top glass overlay:** search/location, map mode, notification, GPS state. Keep compact and allow map gestures.
- **Context chips:** 1-km Find, places category, district filter, weather toggle, hazard legend; only if relevant to the current flow.
- **Main map:** markers for location/place/route/user-consented friends only; cluster many pins; show selected place or hazard detail bottom sheet.
- **Bottom controls:** primary action(s) and a collapsible route/plan summary above bottom navigation; respect SafeArea and keyboard.
- **Accessibility:** 44–48 logical pixel touch targets, readable contrast, semantic marker lists, full Tamil/ Sinhala/English text support and low-width modes.
- **Map states:** initial/permission denied/permission permanently denied/GPS off/key missing/tiles unavailable/no results/rate limited/expired trip. All must offer clear next steps, not misleading “live” claims.
- **Security:** restrict Android Maps SDK key by Android package and signing SHA; server-only API keys for Routes/Places/Geocoding when using protected web service credentials; never embed Gemini/Resend/ Cloudinary secrets in Dart.

Android Emulator QA: valid SDK setup and network access are required. Chrome/Web map support should be implemented separately only if it becomes an explicitly supported platform. A Chrome success or failure does not prove Android emulator behavior. Keep map-independent navigation functional when tiles are unavailable.

5. Recommended professional color/theme system

These are proposed theme tokens, not color values explicitly finalized by the BRD. The original screenshots repeat a bright waterfall background, which reduces map contrast. Keep the original branded scenic **Home/Auth** where desired; use primarily color-driven gradient/glass surfaces for module and map overlays.

Context	Background	Main Accent	Secondary / UI
Auth / Common Home	`#071725` deep scenic navy	`#00B4D8` cyan	`#F8FAFC` foreground
Traveler	`#041C33` deep ocean blue	`#0077B6` blue	`#00B4D8` cyan / `#90E0EF` soft edge
Hunter	`#051F20` dark forest	`#235347` leaf green	`#8EB69B` sage / `#DAF1DE` light
Game	`#170B32` night violet	`#8339B3` purple	`#9D4CCC` lilac / green reserved for success
Profile / Friends	`#061B2D` blue-black	`#00B4D8` cyan	slate blue outlines
Wallet / Coins	`#071A2B` navy	`#00B4D8` cyan	`#F5C451` level/reward gold
Glass / Status	`#0F172A` with variable opacity	high-contrast white	approved green, pending amber, rejected red

Implementation: Put every token in `core/theme/wayhunter_palette.dart` plus new `core/theme/wayhunter_design_tokens.dart`; material color schemes in `app/app_theme.dart`; share `GlassPanel`, `NeonButton`, `ModuleScaffold`, `AppThemedBackground`. Do not hardcode page-specific colors in 40 different Screens. Decorative gradients and background assets may change, but **Do not recolor baked-in Home banner PNGs in place**; replace asset variants if a new banner palette is approved.

6. GitHub-aligned proposed folder structure

Keep the **exact existing folder spelling** **`traveller`** (not `traveler`), `games`, `hunter`, `wallet/coins` during migration. Existing `views/` naming is retained (not `screens/`). New directories are **additive**; no mass renaming.

```
lib/
├── main.dart                              [KEEP]
├── app/
│   ├── app.dart                           [KEEP]
│   ├── app_router.dart                    [MODIFY: additive routes/guards]
│   └── app_theme.dart                     [MODIFY: central themes]
├── core/
│   ├── constants/                         [KEEP]
│   ├── mock/                              [KEEP demo-only; retire in production]
│   ├── network/                           [ADD: api_client, auth_interceptor, failures]
│   ├── auth/                              [ADD: secure session/token storage]
│   ├── maps/                              [ADD: maps_facade, marker models, controllers]
│   ├── services/                          [MODIFY: maps, camera, GPS, permissions]
│   ├── realtime/                          [ADD: signalr transport, lifecycle, reconnection]
│   ├── storage/                           [KEEP local demo; ADD secure draft/cache]
│   ├── theme/                             [MODIFY palette; ADD tokens]
│   └── widgets/                           [MODIFY shared map/glass/navigation UI]
├── shared/
│   ├── domain/entities/                   [ADD: user, place, geo_point]
│   └── presentation/widgets/              [ADD only if reused widely]
└── features/
    ├── auth/                              [KEEP UI, ADD API repository]
    ├── home/                              [KEEP UI, ADD ads/live content]
    ├── friends/                           [KEEP UI, ADD domain+data]
    ├── traveller/                         [KEEP screens; MODIFY maps and trip API]
    ├── hunter/                            [KEEP screens; MODIFY evidence+approval]
    ├── games/                             [KEEP screens; MODIFY coins/expiry/approval]
    ├── leaderboards/                      [ADD: 3 module boards]
    ├── profile/                           [KEEP UI; ADD domain+data]
    ├── wallet/                            [KEEP coins UI; ADD API repository/ledger DTO]
    ├── notifications/                     [ADD: history, FCM, hazard event UI]
    ├── places/                            [ADD or share within traveller/hunter]
    ├── weather/                           [ADD optional overlay feature]
    └── advertisements/                    [ADD campaigns/banners as needed]
```

Per-feature shape (not every tiny feature needs all folders immediately):

```
features/traveller/
├── data/
│   ├── datasources/traveller_remote_datasource.dart    [ADD]
│   ├── dtos/trip_dto.dart                               [ADD]
│   ├── mappers/trip_mapper.dart                         [ADD]
│   └── repositories/api_traveller_repository.dart       [ADD]
├── domain/
│   ├── entities/                                        [KEEP]
│   ├── repositories/                                    [KEEP]
│   └── usecases/                                        [ADD for workflows]
└── presentation/
    ├── providers/                                       [KEEP/EXTEND]
    ├── viewmodels/                                      [KEEP/EXTEND]
    ├── views/                                           [KEEP existing UI]
    └── widgets/                                         [MODIFY map/widgets]
```

7. Exact existing files: KEEP, MODIFY, ADD

Priority	Existing file(s)	Change
KEEP	`lib/main.dart`, `lib/app/app.dart`	Riverpod root, app bootstrap; no rewrite needed
MODIFY	`lib/app/app_router.dart`	Preserve routes; add scoped leaderboards/notifications and typed deep-link parameters
MODIFY	`lib/app/app_theme.dart`, `lib/core/theme/wayhunter_palette.dart`, `wayhunter_module_theme.dart`	New token-driven palettes/backgrounds, consistent typography

Priority	Existing file(s)	Change
MODIFY	`lib/core/services/maps_service.dart`	Remove Android-only demo map assumptions; proper availability, protected provider API contracts
MODIFY	`lib/features/traveller/presentation/widgets/traveller_island_map.dart`	Genuine Traveler landing map and filtered approved place pins
MODIFY	`lib/features/traveller/presentation/widgets/traveller_map.dart`	Active map, route, valid hazards, opt-in friend markers, fallback
MODIFY	`lib/features/traveller/presentation/widgets/trip_route_map.dart`, `trip_location_picker.dart`	Real place search + selected location + road route preview
MODIFY	`lib/features/hunter/presentation/views/hunter_compact_map.dart`	Detected district + approved discoveries + zoom/filter
MODIFY	`lib/features/games/presentation/views/game_current_district_screen.dart`	District selector optional; do not expose hidden game locations
MODIFY	`lib/features/games/domain/entities/game_models.dart`, `game_view_model.dart`, `game_flow_screens.dart`	Remove obsolete HP/RC; add Admin review status, publication/expiry and Coins source display
MODIFY	`lib/features/hunter/presentation/viewmodels/hunter_view_model.dart`, `hunter_evidence_screens.dart`	Submit to Admin API, pending/approved/rejected, real protected evidence upload
MODIFY	`lib/features/friends/presentation/views/friends_screens.dart`, `friends_view_model.dart`	Single Friends context, app deep link, permissioned contact matching backend
MODIFY	`lib/features/wallet/coins/coin_wallet_view_model.dart`, `coin_wallet_screens.dart`	Keep Coins visual design; API source of truth, pending reservations, issued vouchers
MODIFY	`lib/features/profile/presentation/views/profile_dashboard_screen.dart`	Module UI tabs, unified Friends, only Coins/Level; settings endpoint wiring
MODIFY	`lib/core/widgets/app_nav_bar.dart`, `home_profile_bottom_nav.dart`, `wayhunter_bubble_bottom_nav.dart`	Remove forbidden Global Leaderboard action on named pages; central nav behavior
ADD	`lib/core/network/`, `lib/core/auth/`, `lib/core/realtime/`, `lib/core/maps/`	Cross-feature services built to typed interfaces
ADD	`lib/features/leaderboards/`, `lib/features/notifications/`	Proper features, not placeholder screens
KEEP (dev)	`docs/flows/`, `docs/original_screens/`, screenshot tests	References and golden-regression baseline

Old mock and `*.bak` files should not be deleted during the initial architecture migration. Avoid breaking GitHub history. Retire obsolete demo routing only after API path and tests are stable.

8. Feature responsibilities and BRD rules

Auth / Home

Registration with immutable server `UserId`; verified OTP and one-time `+100 Coins` account activation, not every reinstall; language Tamil/Sinhala/English, just-in-time permissions, secure session refresh; Home greeting, 3 image-based module cards, sponsored banner rotation, no global leaderboard shortcut. Use backend-verified active status and prevent a suspended/deleted account from submitting or spending.

Friends / Profile

One Friends page for Profile and Trip Add Friends, with Followers/Following, Follow Requests, Discover, Invite via OS chooser, device Contacts matching only after consent, Profile Public/Private and blocks. Following is **not** trip membership. An accepted trip invitation is **not** GPS consent. Keep profile Games/Posts/Trips/Hunt/My Wallet tabs, settings, language, security, account deletion and logout.

Traveler / Places / AI / Safety

Trip Draft with ordered start/stops/end, budget/vehicle/headcount; copy suggested plans without mutating original; server-mediated Gemini trip plan is editable and estimates labelled; only actual **Start → Finish** generates a completed trip and increases Traveler leaderboard count. Find 1-km on explicit tap. Active route live map shows only authorized friends and active hazards; create hazard with live evidence, label as community/unverified until authorized verification; finish stops location sharing.

Hunter

Evidence sequence: current GPS district + accuracy → live guided 360-video/normal photo (no Gallery) → duplicate candidate screening 200m → submit UserId/Name → Admin Pending → Approve/Reject → owner sees My Hunts state. `+200 Coins` and publishing only after eligible approval, one time. Private raw evidence remains private after publication.

Games

Creator can use Gallery for *clue* media, not for player proof; Game Create/Submit → Manual Admin Review → Published active (creator earns `+100 Coins`) → expires one month after defined clock anchor (requires confirmation) → hidden public and archived in creator My Games. Player sees photo/text clue and timer; capture **live camera proof**, submit final result → Admin approval → `+100 Coins` only for eligible completed result. Creator cannot play own game. First-winner vs multi-winner rule unresolved; UI must not promise 100 Coins to every finisher until decided.

Wallet / Leaderboards / Notifications

Coins only: `Available`, `Reserved`, `LifetimeEarned` are balances/metrics, not separate currencies. Level star + straight bar for each 2,000 lifetime earned Coins. `10,000 Coins = LKR 100` is reference, not guaranteed cash-out. Redemption Pending leaves visible balance un-debited while **server reserves spendability**; Approve finalizes debit once, Reject releases hold. Voucher code/QR only after real allocation. Game rank uses eligible Game Coins; Hunter rank eligible Hunter Coins; Traveler rank valid completed trip counts. FCM + in-app history, approval statuses, trip reminders without auto-start, hazard popups during active trips.

9. Flutter ↔ backend interface mapping (proposed contracts)

Endpoint names are **architecture proposals**, not verified existing API endpoints. Confirm Swagger/OpenAPI against actual ASP.NET Core backend before implementation.

| Feature | Proposed API / event | Client responsibility | |---|---|---| | Auth | `POST /api/auth/register`, `/verify-otp`, `/login`, `/refresh`, `/logout` | Persist secure session; refresh + correct route guard | | Current user | `GET /api/users/me`, `PATCH /api/users/me`, `/settings` | Profile/consent, initial name and UserId | | Friends | `GET /api/friends`, `POST /api/follows/requests`, `/respond`, `/contacts/match` | Follow requests / permissioned matches / invite UI | | Trip | `GET/POST /api/trips`, `/trips/{id}/start`, `/finish`, `/invites` | Draft, actual start/end, no duplicate finish | | Maps/Places | `GET /api/places/search`, `/maps/route`, `/places/approved` | Search + road geometry + verified pins, provider fallbacks | | Hazards | `POST /api/hazards`, `GET /api/hazards/near-route` | Report and live safe overlays | | Hunter | `POST /api/hunter/submissions`, `GET /api/hunter/mine`, `GET /api/hunter/submissions/{id}` | Pending/Approved/Rejected, private evidence refs | | Games | `POST /api/games`, `GET /api/games`, `POST /api/games/{id}/attempts`, `/proof` | Admin review states, expiry, timers | | Wallet | `GET /api/wallet`, `GET /api/wallet/ledger`, `POST /api/redemptions` | Coins authority, pending hold and voucher states | | Voucher | `GET /api/vouchers/mine`, `GET /api/vouchers/{id}` | Only genuine issued QR/code, usage/expiry | | Leaderboards | `GET /api/leaderboards/{traveler\|hunter\|game}` | View sorted eligible aggregates and own position | | Notifications | `GET /api/notifications`, `POST /api/devices/fcm` | History, permission, read receipts, remote push token | | SignalR | authenticated `/hubs/wayhunter` (proposed) | Approval, wallet, friend request, trip/hazard status; refetch API |

Use typed DTOs, validation, cancellation, `AsyncValue` loading/error/empty, short-lived auth headers and safe retry/backoff. **Never retry non-idempotent POST blindly**; use operation/request IDs and backend idempotency keys. For media use private signed upload authorization and metadata completion rather than public static URLs.

10. Riverpod state patterns

- **View → ViewModel:** `ConsumerWidget` observes `AsyncValue` or typed immutable state; form controllers hold local draft; ViewModel owns submit/refresh commands.
- **Domain:** typed `Trip`, `Game`, `HunterSubmission`, `CoinWallet`, `Voucher`, `User`, `GeoPoint`, plus enums, and Repository Interfaces.
- **Data:** `RemoteDataSource` converts JSON to DTO; Mapper converts DTO to domain; Repository coordinates remote data and safe cache.
- **Realtime:** SignalR connects on sign in, reconnects and subscribes with authorization; event triggers provider invalidation, not a direct wallet debit.
- **Draft resilience:** Preserve Create Trip form when navigating to Friends/Place Picker; persist only drafts permitted by privacy policy, not live GPS and tokens in plain SharedPreferences.
- **Media:** use camera/guided capture dedicated flow, stream/upload progress, temporary secure files, permission denied/retake/cancel; hide Gallery on Hunter and Player Proof; allow permitted upload on Creator clue photos.

```
// Illustrative only. Backend is the authority for Coins.
final coinWalletProvider = AsyncNotifierProvider<
    CoinWalletNotifier, CoinWallet>(CoinWalletNotifier.new);

class CoinWalletNotifier extends AsyncNotifier<CoinWallet> {
  @override
  Future<CoinWallet> build() async => ref.read(walletRepositoryProvider).fetchWallet();

  Future<void> refreshWallet() async {
    state = await AsyncValue.guard(
      () => ref.read(walletRepositoryProvider).fetchWallet(),
    );
  }
}
```

11. Flutter dependencies and ownership

Present in inspected GitHub `pubspec.yaml`: `flutter_riverpod`, `go_router`, `google_maps_flutter`, `geolocator`, `http`, `camera`, `image_picker`, `shared_preferences`, `intl`, `uuid`, `qr_flutter`, `fancy_bottom_navigation_plus`. **Potential additions (only when implementing feature):** `flutter_secure_storage` for session tokens; `firebase_core` and `firebase_messaging` for FCM; `share_plus` for native share sheet; contacts package with explicit consent for My Contacts; SignalR-compatible client after backend transport is established; `flutter_localizations` / `intl` generated resources. Check compatibility before any `pubspec` change; the list is not a claim these are installed.

Do not force an unrelated bottom-navigation package just to match a static screenshot. Keep the existing navigation widget and replace only where it materially improves tested behavior.

12. Tests, migration steps and sign-off

Tests (add or extend existing suites)

Suite	Required proof
Auth	signup/OTP once bonus +100; session, consent and logout behavior
Visual golden tests	390×844, 320×640/568, Tamil/Sinhala overflow, profile tabs, bottom nav, background themes
Maps on Android emulator	tiles configured, place search, map tap, re-center, markers, road route/ETA only when provider returns, offline/failure fallback
Trip	start/finish actual events; draft preserved after Friends; 1-km Find; consented member markers; active hazard pop-up
Hunter	camera-only no gallery, 360-step, 200m candidate, Admin pending/approve/reject, +200 exactly once
Games	creator/gallery vs proof/camera, no HP/RC, Admin double review, +100 credits, one-month public expiry, creator archive
Wallet	available/reserved/earned, ≥10k provisional boundary, pending no final debit, duplicate hold prevention, approve/reject/QR issue
Real-time & security	authorized SignalR subscriptions, declined permission, token expiration, duplicate retry, no secret API keys in Dart
Regression	`flutter analyze`, `flutter test`, screen automation flows, Android smoke testing

Phased non-destructive migration:

1. **Freeze UI references:** preserve screenshots/flow mapping; create branch `feature/frontend-architecture-brd-v2` locally before coding (only a suggestion; not created here).
2. **Centralize themes and map foundation:** tokens and real-map components, leaving old routes intact.
3. **Introduce Auth/Core API/secure storage:** mock and live repositories interchangeable via Provider overrides, but never allow demo rewards in Production.
4. **Migrate Traveler map-first views + Friends:** route picker, road routes, Find Nearby, Trip Draft/Start/Finish, safety/consent.
5. **Migrate Hunter/Game approval and Coins source:** eliminate legacy HP/RC in domain and user UI, add one-time verified reward events.
6. **Migrate wallet, leaderboards and notifications:** backend-driven state, issuer code, in-app+FCM history, schedule.
7. **Run test matrix and release checks:** API contract, field safety, map key restrictions, data migration, Android acceptance, owner sign-off.

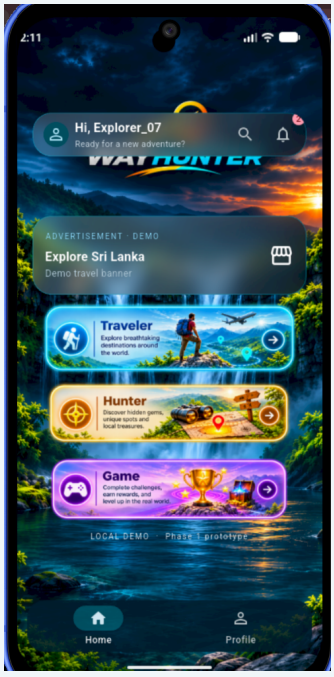
Owner decision log (do not silently invent)

1. Confirm **new color palette** and whether the waterfall scenic image remains on Auth/Home only or is kept as a faint module overlay.
2. Final bottom-navigation fourth slot / removal of global Leaderboard shortcut.
3. Game 1-month clock anchor and what happens when an attempt is underway at expiry.
4. Game single-first-winner vs multi-eligible-completion Rewards.
5. Minimum redemption exactly 10,000 or strictly above 10,000 Coins; payment vs voucher route and approved payout contract.
6. Hazard pop-up radius/age/cooldown; location refresh frequency and privacy retention.

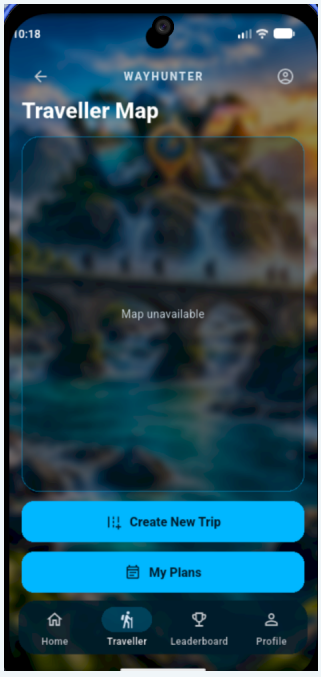
Final recommendation: KEEP existing screens/flow and folder names → MODIFY map surfaces and centralized color system → ADD backend API repository boundaries, three module leaderboards and production notification services. This is an architecture deliverable; it does **not** alter existing Flutter GitHub code or imply that APIs are already connected.

Appendix — ZIP Visual Audit (selected 6 of 42)

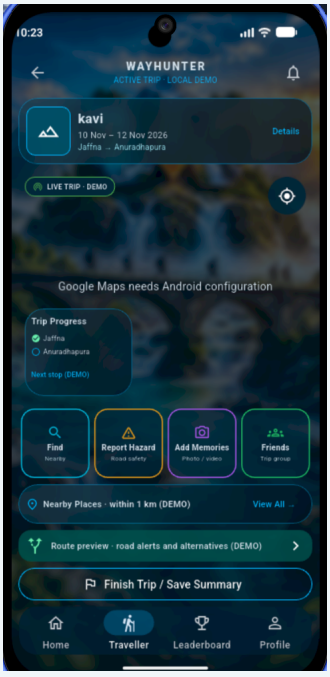
These are user-supplied existing UI screenshots; they show current references and DEMO placeholders, **not** the redesigned map screens.



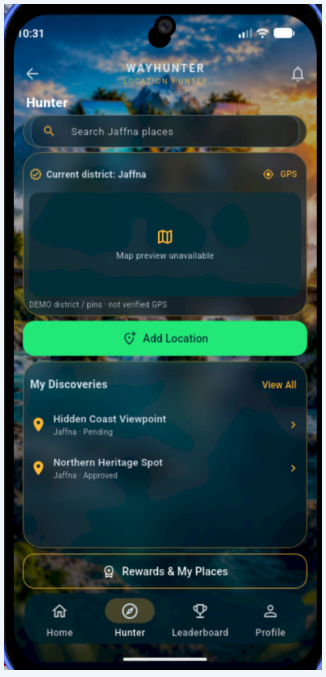
Current Home (3 module cards)



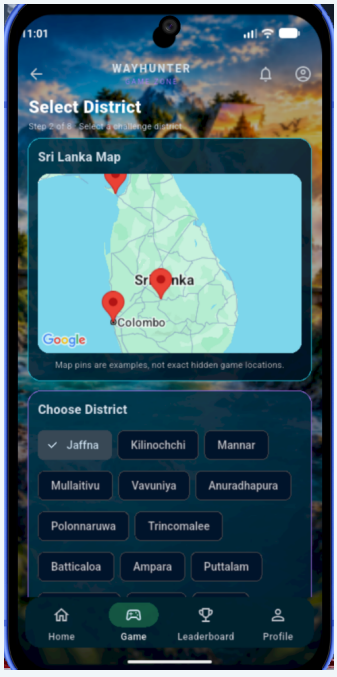
Traveler: map unavailable



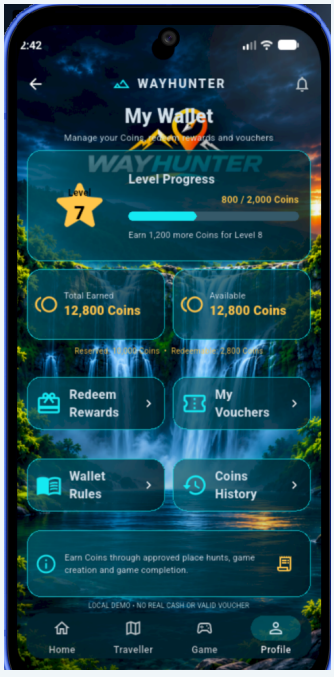
Active Trip: map setup needed



Hunter: demo district map



Game: optional district selector



Wallet: Coins + star progress